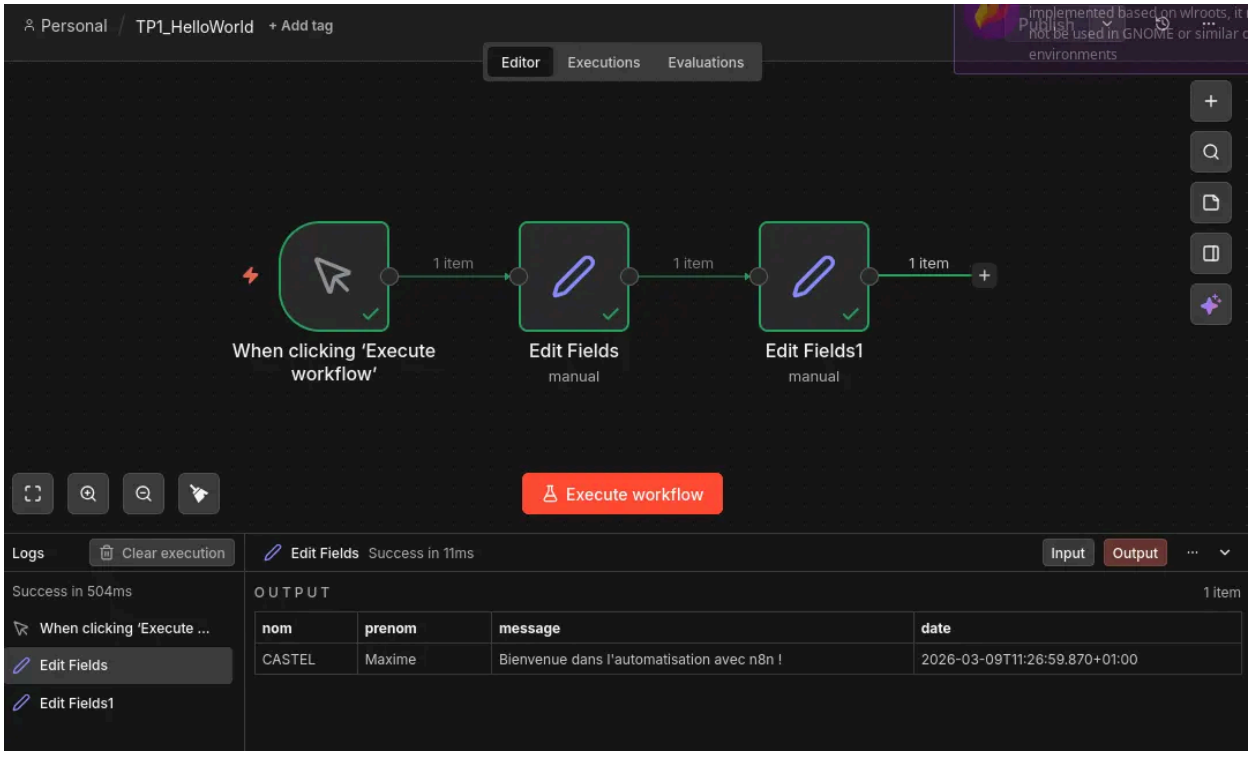
TP jour 1 - Introduction a l'I.A et Automatisation

TP 1



- Que se passe-t-il si vous exécutez le workflow plusieurs fois ?
  - Seul le timestamp change
- Comment les données circulent-elles entre les nodes ?
  - Elles circulent sous forme d'objets JSON

TP 2

Schedule Trigger

Parameters | Settings | Execute step

This workflow will run on the schedule you define here once you publish it.For testing, you can also trigger it manually: by going back to the canvas and clicking 'execute workflow'

Trigger Rules

Trigger Interval: Days

Days Between Triggers: 1

Must be in range 1-31

Trigger at Hour: 9am

Trigger at Minute: 0

Add Rule

I wish this node would...

OUTPUT

No trigger output

Test this trigger

or set mock data

Search previous nodes' fields

Schedule Trigger — 1 item

timestamp	2026-03-09T12:00:36.213+01:00
Readable date	March 9th 2026, 12:00:36 pm
Readable time	12:00:36 pm
Day of week	Monday
Year	2026
Month	March
Day of month	09
Hour	12
Minute	00
Second	36
Timezone	Europe/Paris (UTC+01:00)

Variables and context

Expression

Anything inside {{ }} is JavaScript. Learn more

```
{
  "date_rapport" : "{{ $now.format('DD/MM/yyyy') }}",
  "heure_rapport" : "{{ $now.format('HH:mm:ss') }}",
  "jour_semaine" : "{{ $now.weekdayLong }}",
  "numero_jour" : "{{ $now.ordinal }}",
  "message" : "Rapport quotidien généré automatiquement"
}
```

Result

Item C — Text | Html | Markdown

```
{
  "date_rapport" : "Mar 9, 2026/03/2026",
  "heure_rapport" : "12:02:35",
  "jour_semaine" : "Monday",
  "numero_jour" : "68",
  "message" : "Rapport quotidien généré automatiquement"
}
```

Code in JavaScript

INPUT — Schema | Table | JSON

Edit Fields — 1 item

date_rapport	Mar 9, 2026/03/2026
heure_rapport	12:07:12
jour_semaine	Monday
numero_jour	68
message	Rapport quotidien généré automatiquement

Schedule Trigger — 1 item

Variables and context

Parameters | Settings | Execute step

Mode: Run Once for All Items

Language: JavaScript

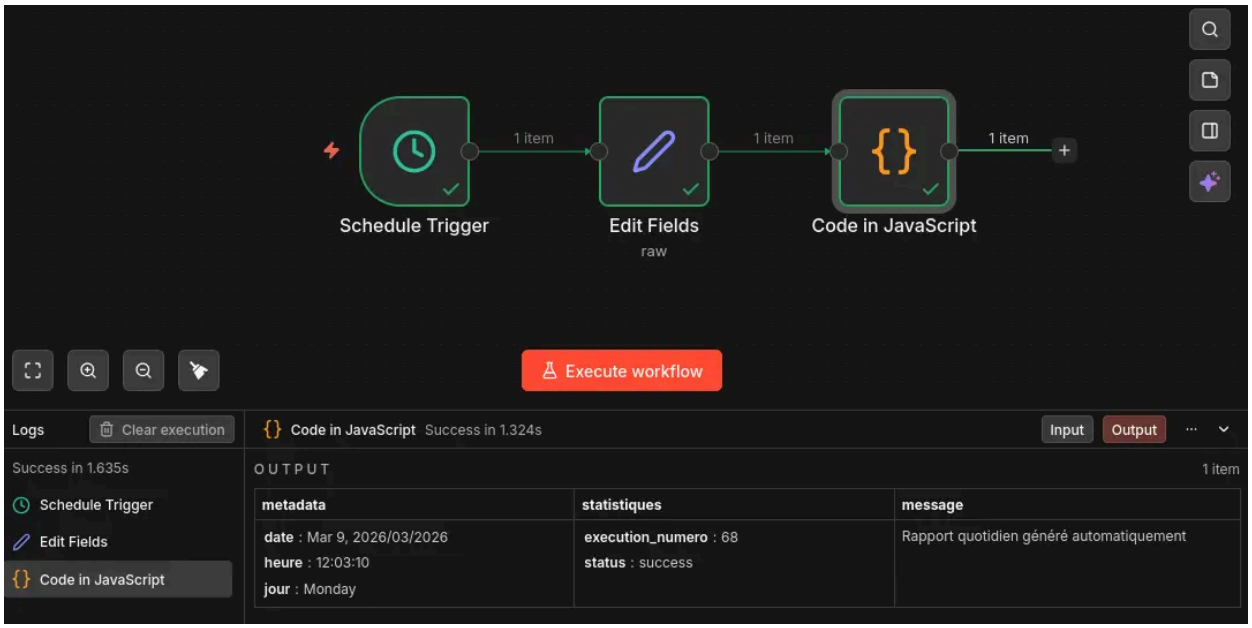
JavaScript — Code | Ask AI

```
// Loop over input items and add a new field called 'myNewField' to the JSON of each one
const rapport = { metadata: { date: $input.first().json.date_rapport, heure: $input.first().json.heure_rapport, jour: $input.first().json.jour_semaine },
statistiques: { execution_numero: $input.first().json.numero_jour, status: "success" }, message: $input.first().json.message }; return { json: rapport };
```

OUTPUT — Schema | Table

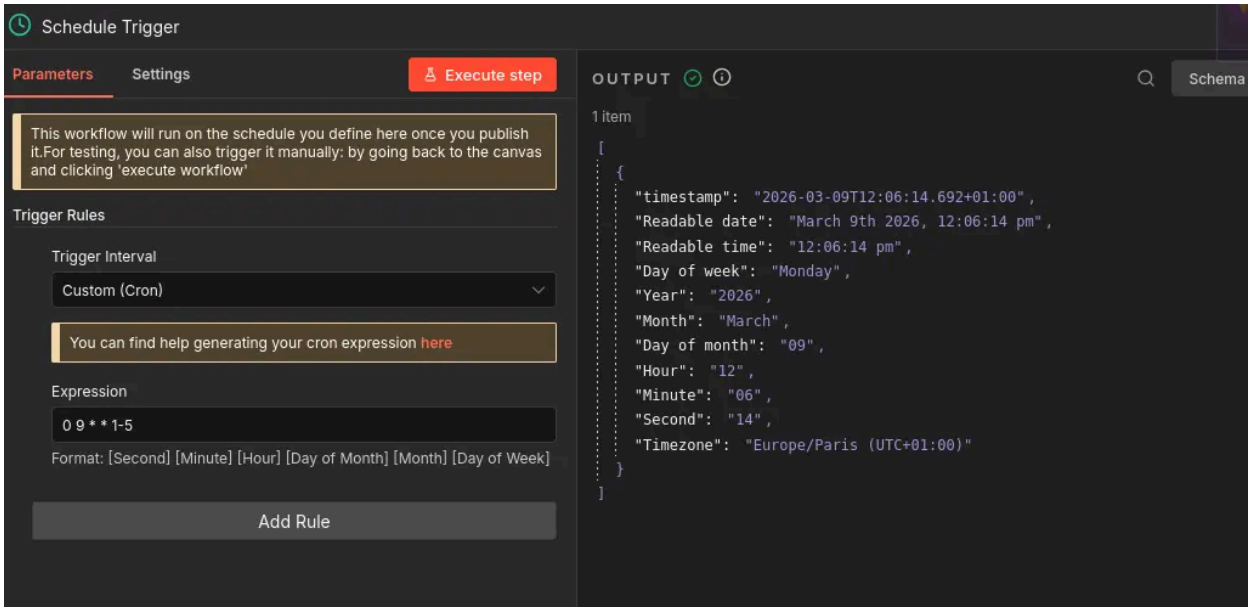
1 item

```
[
  {
    "metadata": {
      "date": "Mar 9, 2026/03/2026",
      "heure": "12:07:12",
      "jour": "Monday"
    },
    "statistiques": {
      "execution_numero": "68",
      "status": "success"
    },
    "message": "Rapport quotidien généré automatiquement"
  }
]
```

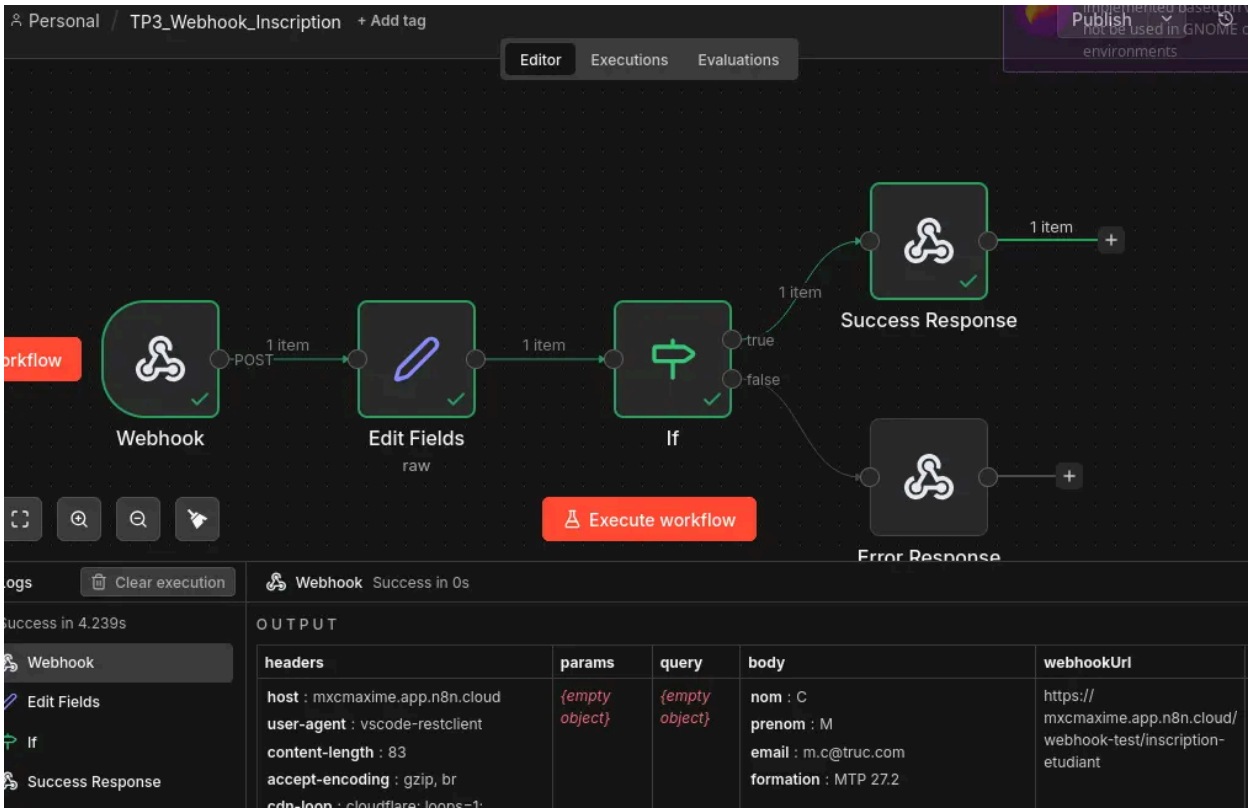


Execution en jour de semaine (Lundi a vendredi)

```
0 9 * * 1-5
```



TP 3



email valide :

```
POST https://mxcmaxime.app.n8n.cloud/webhook-test/inscription-etudiant
Content-Type: application/json

{
    "nom" : "C",
    "prenom" : "M",
    "email" : "m.c@truc.com",
    "formation" : "MTP 27.2"
}
```

```
POST https://mxcmaxime.app.n8n.cloud/webhook-tes
Content-Type: application/json

{
"nom" : "C",
"prenom" : "M",
"email" : "m.c@truc.com",
"formation" : "MTP 27.2"
}
```

```
HTTP/1.1 200 OK
Date: Mon, 09 Mar 2026 11:30:27 GMT
Content-Type: application/json; charset=utf-
Transfer-Encoding: chunked
Connection: close
CF-Ray: 9d99bf6b8f9ecac8-MRS
CF-Cache-Status: DYNAMIC
ETag: W/"38-LhWtY2X/Vbne49XblAMd5dAAhqg"
Server: cloudflare
Strict-Transport-Security: max-age=15552000;
deSubDomains
Vary: Accept-Encoding, Accept-Encoding
Nel: {"report_to":"cf-nel","success_fractior
0,"max_age":604800}
Report-To: {"group":"cf-nel","max_age":60480
points":[{"url":"https://a.nel.cloudflare.co
rt/v4?s=pjWDC56KA4rmcLPceBAp9wsF5Ixm4LvhYBvf
IDOwLIAbd76E1LirBYj5iNajmONZFM4Rqr7sScD3Y3J4
ZRRZUGE9BN%2BB2Lokk7wT83o%3D"}]}
content-security-policy: sandbox allow-downl
llow-forms allow-modals allow-orientation-lo
ow-pointer-lock allow-popups allow-presentat
low-scripts allow-top-navigation-by-user-act
n allow-top-navigation-to-custom-protocols
referrer-policy: same-origin
x-content-type-options: nosniff
x-frame-options: SAMEORIGIN
Content-Encoding: gzip

{
  "status": "valide",
  "message": "Inscription enregistrée"
}
```

email invalide :

```
POST https://mxcmaxime.app.n8n.cloud/webhook-tes
Content-Type: application/json

{
"nom" : "C",
"prenom" : "M",
"email" : "m.ctruc.com",
"formation" : "MTP 27.2"
}
```

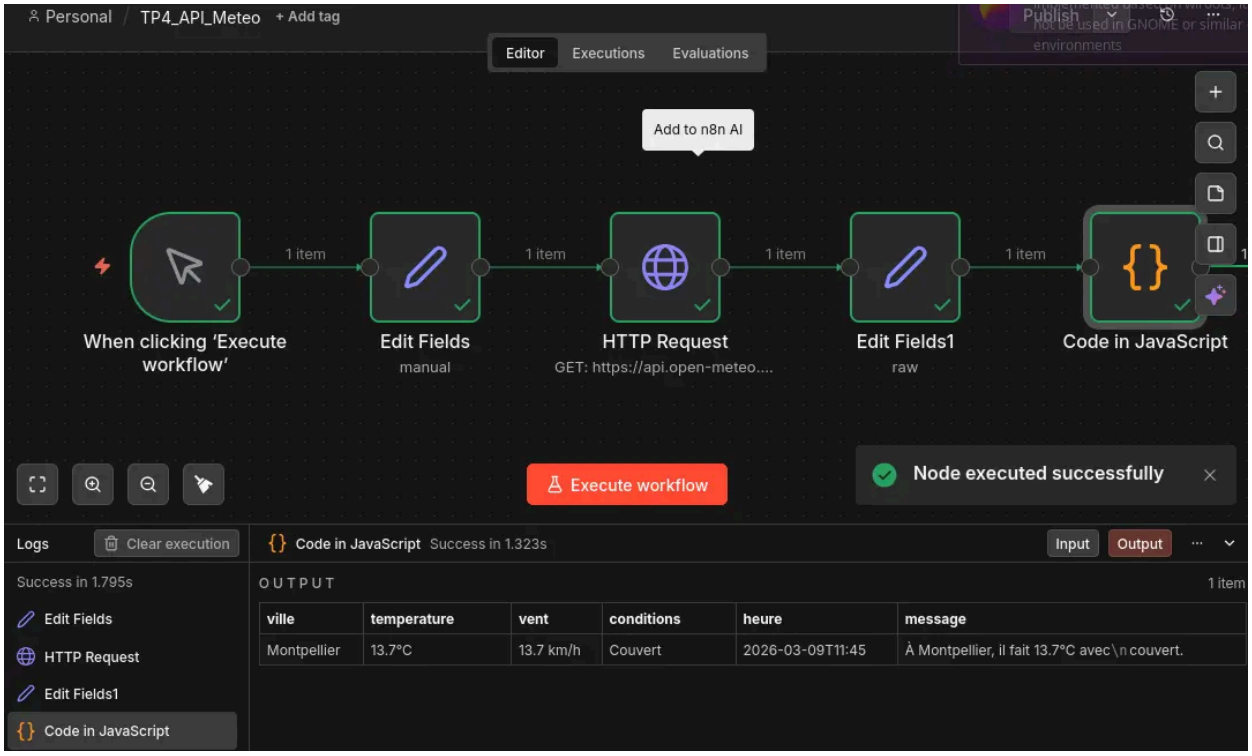
```
Date: Mon, 09 Mar 2026 11:31:36
Content-Type: application/json;
Content-Length: 46
Connection: close
CF-Ray: 9d99c1171edf49f0-MRS
CF-Cache-Status: DYNAMIC
ETag: W/"2e-M2nKqSvm1PZcjTrQ2xIc
Server: cloudflare
Strict-Transport-Security: max-a
SubDomains
Vary: Accept-Encoding, Accept-En
Nel: {"report_to":"cf-nel","succ
ax_age":604800}
Report-To: {"group":"cf-nel","ma
ints":[{"url":"https://a.nel.clo
4?s=jpeUcnuakEd8YrMs%2B5FtNrPFRs
OP%2BK77f7Ibatwr4q4DuH8jukYL7NzB
vVYy%2BdunvyNN30xc5fGNE%3D"}]}
content-security-policy: sandbox
ow-forms allow-modals allow-orie
ointer-lock allow-popups allow-p
ripts allow-top-navigation-by-us
top-navigation-to-custom-protoco
referrer-policy: same-origin
x-content-type-options: nosniff
x-frame-options: SAMEORIGIN

{
  "status": "erreur",
  "message": "Email invalide"
}
```

- Quelle est la différence entre un trigger manuel et un webhook ?
  - Un trigger manuel doit être déclenché a la main en cliquant sur un bouton tandis qu'un webhook est déclenché lors d'une requête http sur un lien avec la possibilité

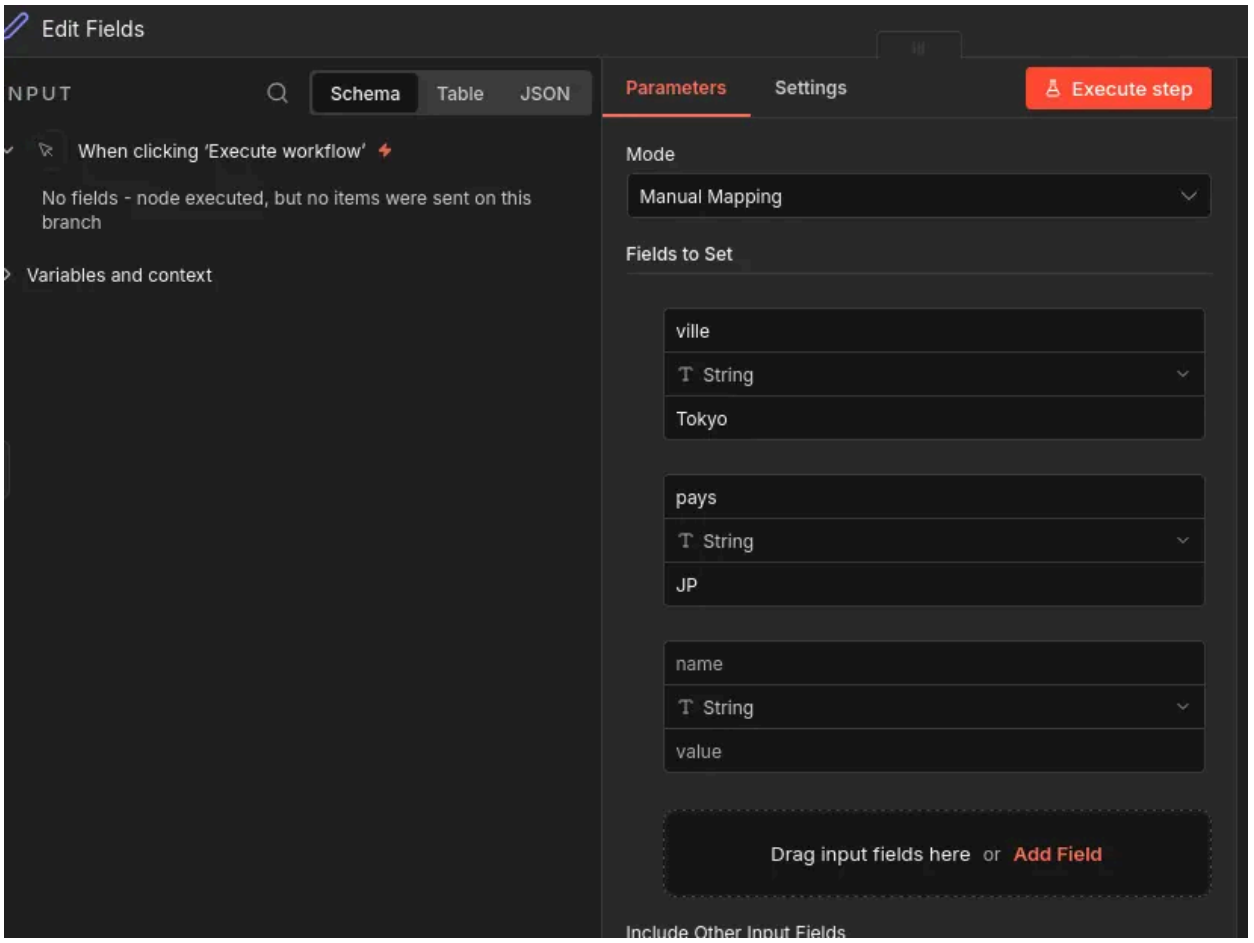
d'y ajouter de la donnée.

- Comment pourriez-vous sécuriser ce webhook en production ?
  - En activant l'authentification utilisant un header avec un JWT par exemple.

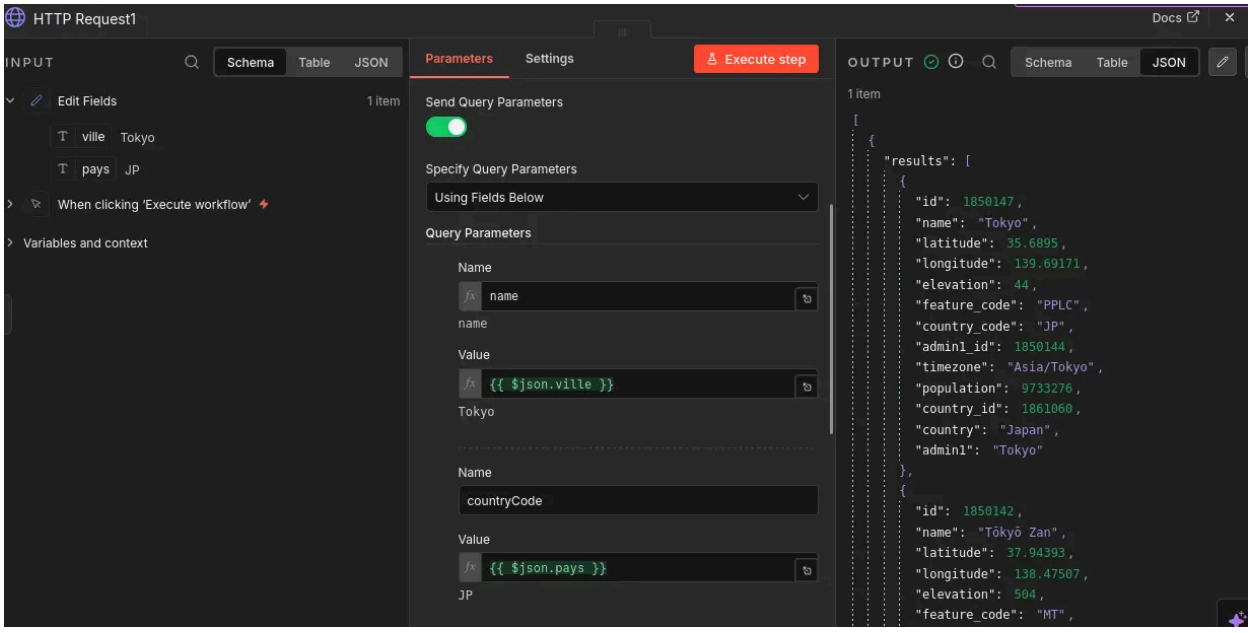
TP 4

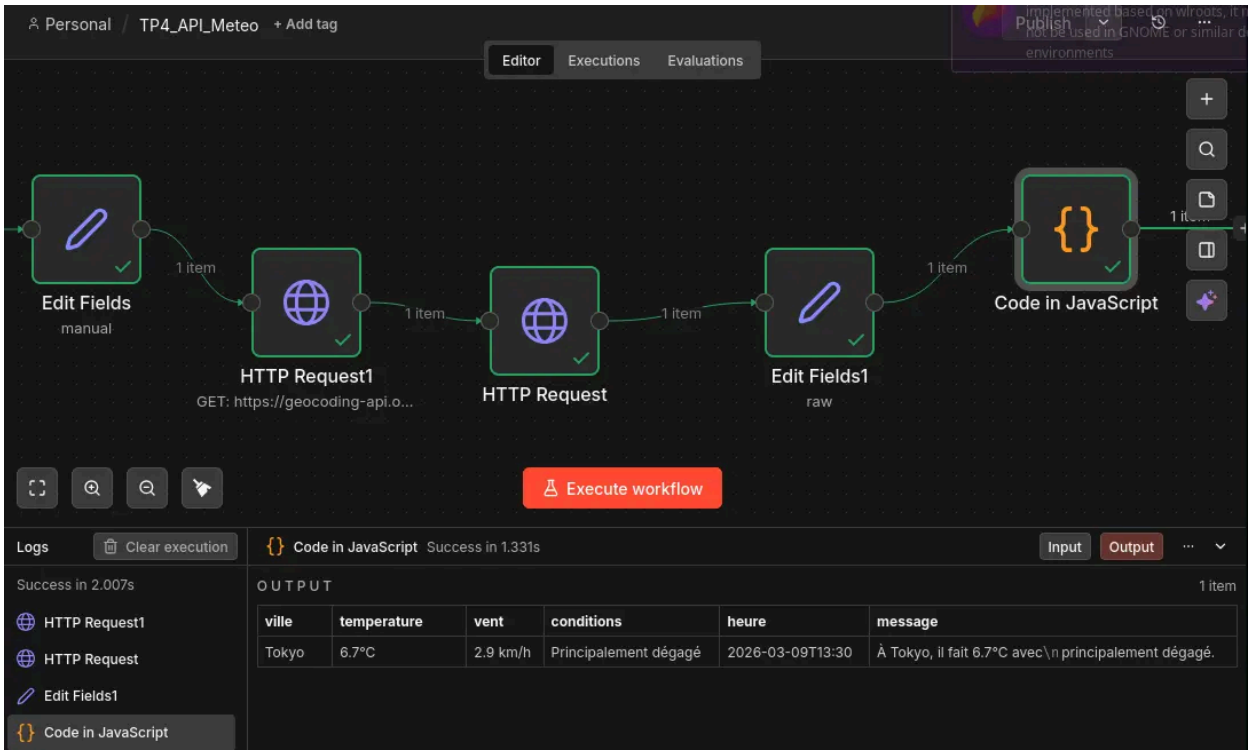


Ajout du choix de la ville:




```
https://geocoding-api.open-meteo.com/v1/search?name=Tokyo
```

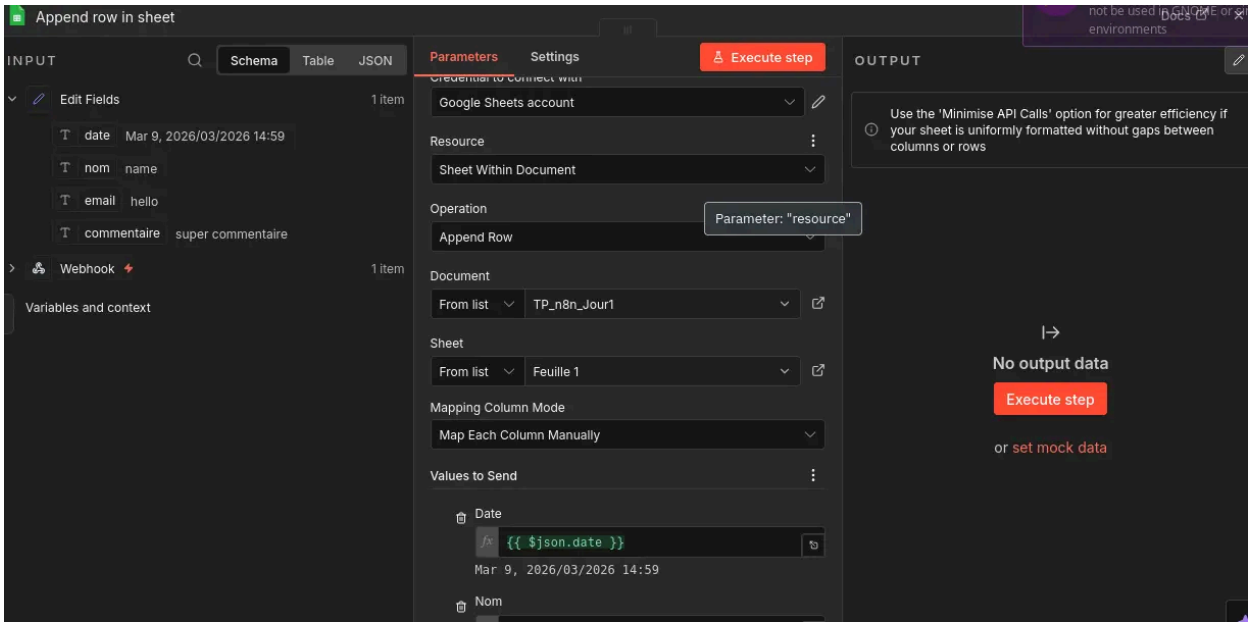




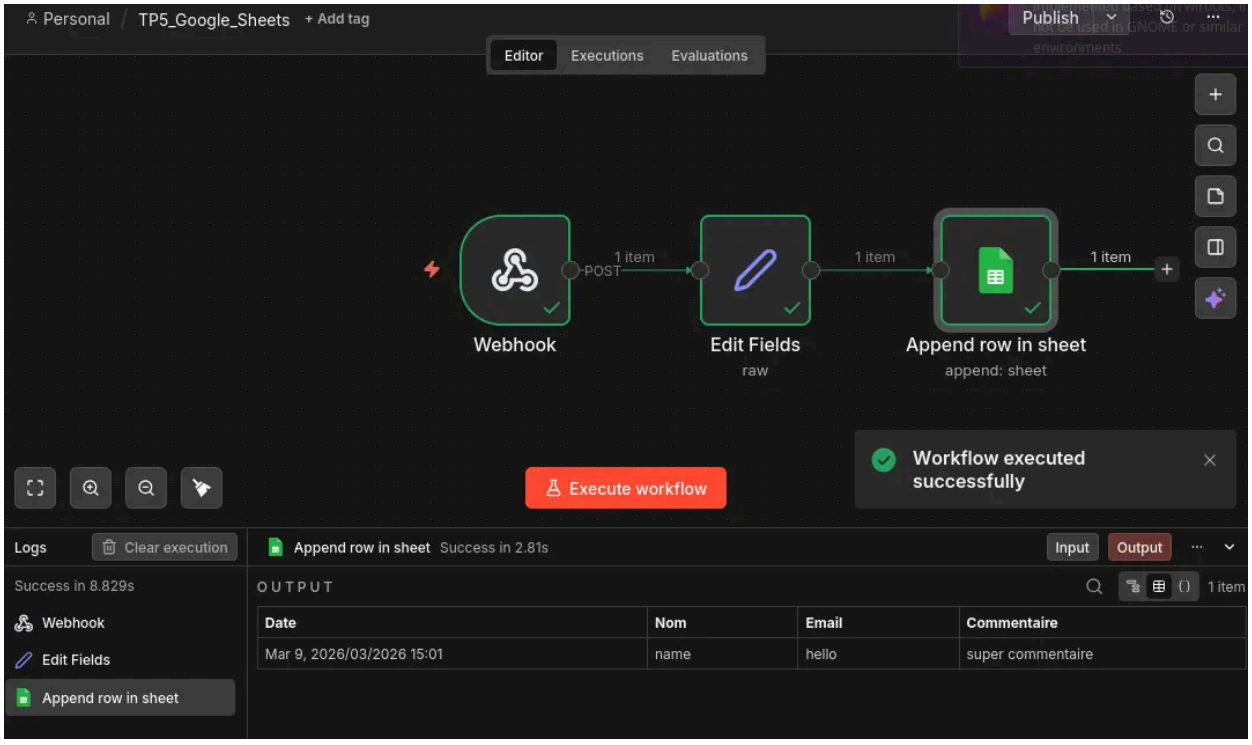
Code final avec l'affichage de "Il fait chaud".

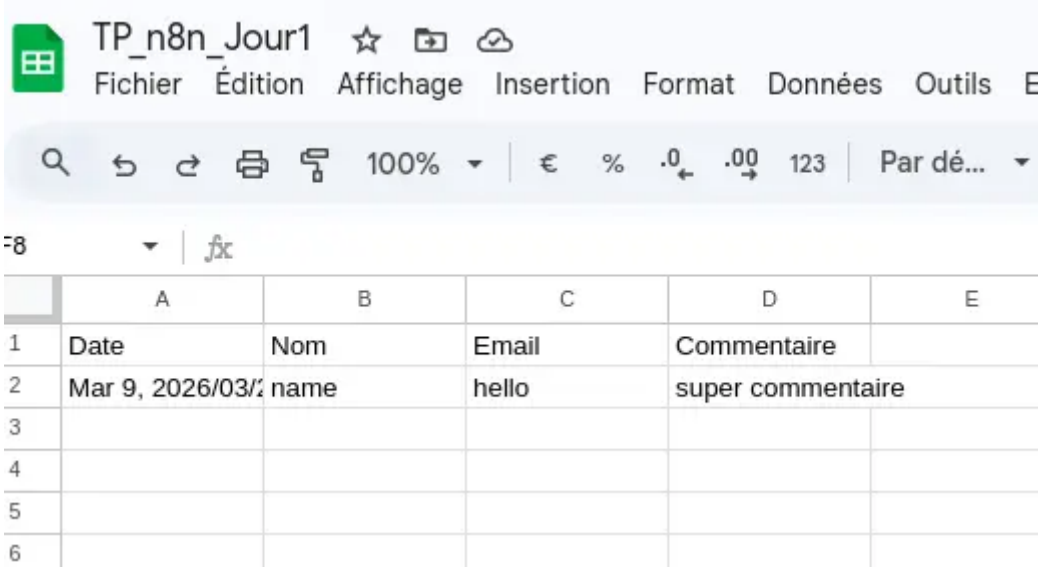
```
const weatherCodes = { 0: "Ciel dégagé", 1: "Principalement dégagé", 2:
"Partiellement nuageux", 3: "Couvert", 45: "Brouillard", 51: "Bruine légère",
61: "Pluie légère", 63: "Pluie modérée", 65: "Pluie forte", 71: "Neige légère",
95: "Orage" };
const ville = $('Edit Fields').first().json.ville
const data = $input.first().json;
const description =
weatherCodes[data.code_meteo] || "Conditions inconnues";
const isHot = Number.parseFloat(data.temperature) > 25 ? "Il fait chaud !" : ""
return {
    json: {
        ville,
        temperature: `${data.temperature}°C`,
        vent: `${data.vitesse_vent} km/h`,
        conditions: description,
        heure: data.heure_releve, message: `À ${ville}, il fait
${data.temperature}°C avec ${description.toLowerCase()}. ${isHot}`
    }
};
```

TP 5

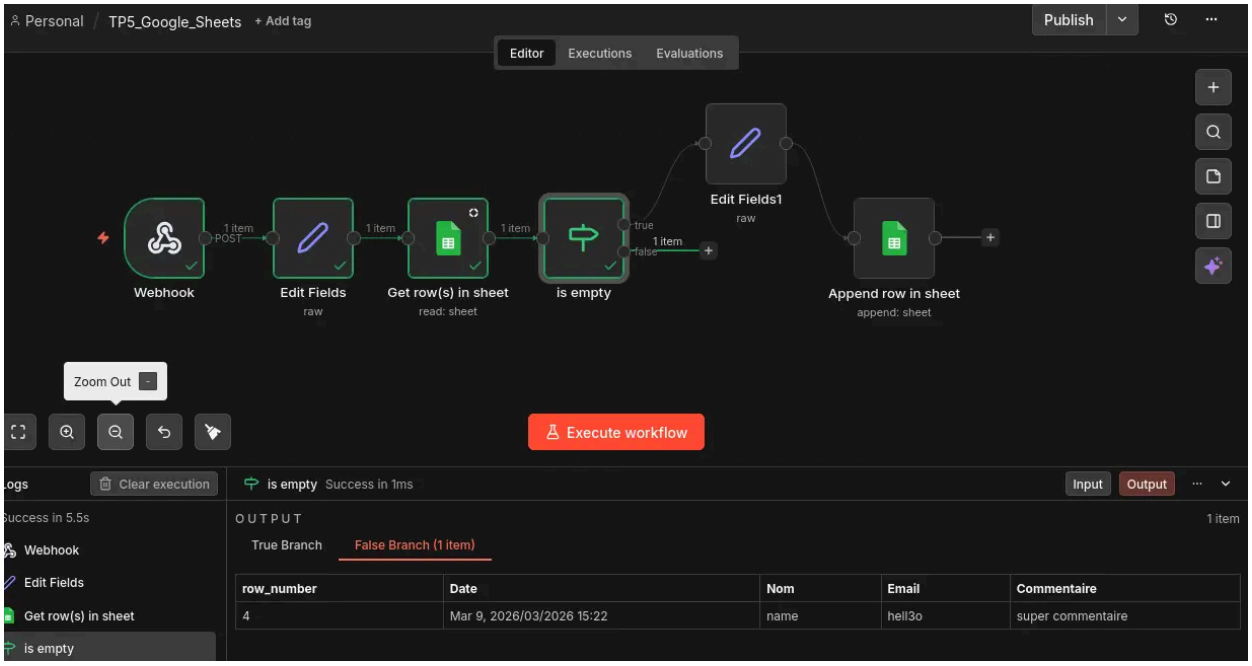


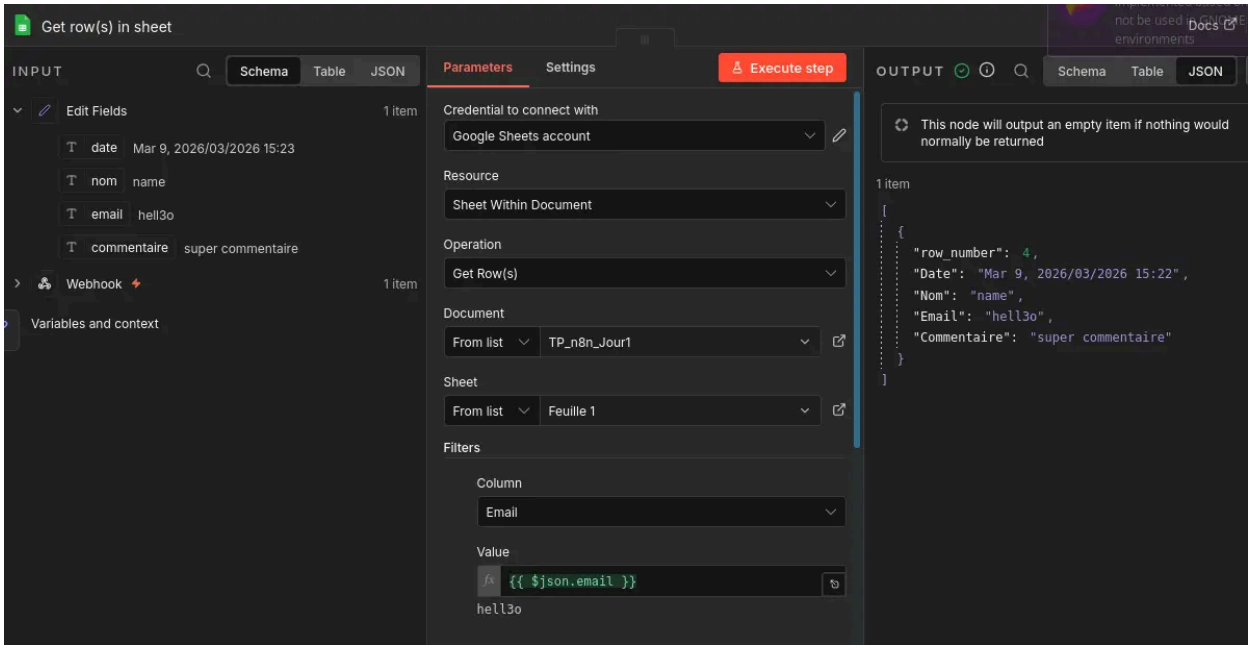

```
POST https://mxcmaxime.app.n8n.cloud/webhook-test/enregistrer-contact
Content-Type: application/json

{
  "nom" : "name",
  "email" : "hello",
  "commentaire" : "super commentaire"
}
```

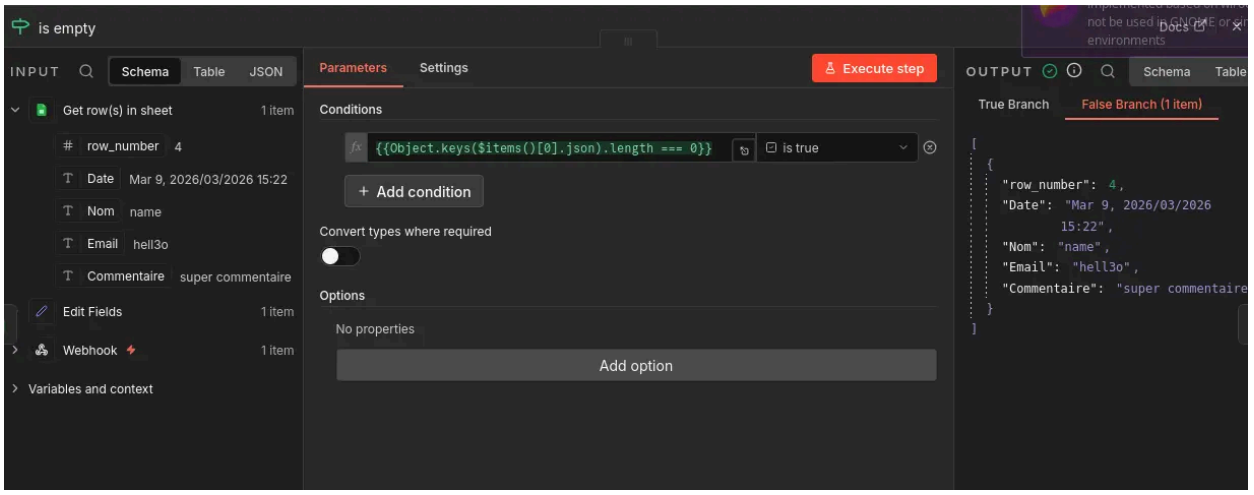




Eviter les doublons :







Exercice de synthèse Jour 1

Mini-projet : Système de notification quotidienne Synthèse

> Objectif : Combiner les concepts appris pour créer un workflow complet

Cahier des charges

1. Un workflow qui s'exécute automatiquement chaque jour à 8h00
2. Récupère la météo de Montpellier
3. Crée un message de motivation aléatoire
4. Combine météo + citation du jour
5. (Optionnel) Envoie le résultat par email ou Slack

Contraintes techniques

- Utiliser au minimum 5 nodes différents
- Inclure au moins une expression n8n
- Gérer le formatage des dates en français
- Structurer les données en JSON propre

Livrables

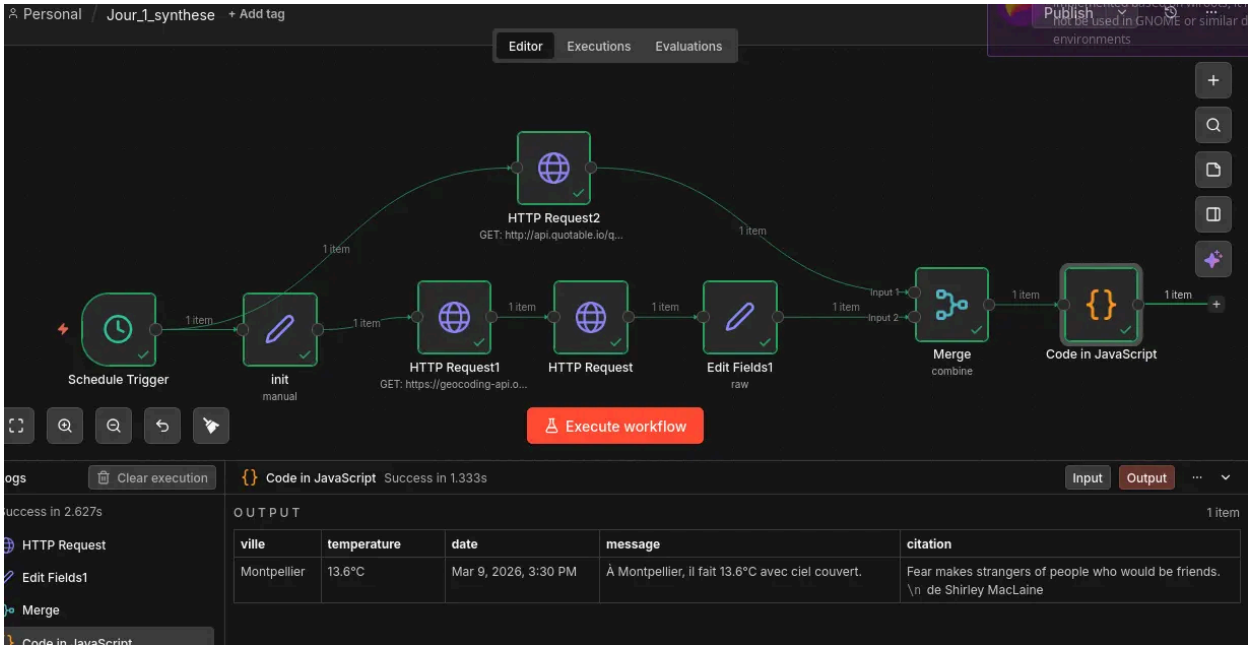
- Workflow fonctionnel et testé
- Capture d'écran du canvas
- Export JSON du workflow

APIs publiques gratuites pour vos tests :

API	URL	Usage
Open-Meteo	https://api.open-meteo.com/	Météo
JSONPlaceholder	https://jsonplaceholder.typicode.com/	Données de test

API	URL	Usage
Random User	https://randomuser.me/api/	Utilisateurs fictifs
Quotes API	https://api.quotable.io/random	Citations aléatoires

Graph



code js:

```
const weatherCodes = { 0: "Ciel dégagé", 1: "Principalement dégagé", 2:
"Ciel Partiellement nuageux", 3: "Ciel Couvert", 45: "Brouillard", 51: "Bruine
légère",
61: "Pluie légère", 63: "Pluie modérée", 65: "Pluie forte", 71: "Neige légère",
95: "Orage" };
const ville = $('init').first().json.city
const data = $input.first().json;
const description =
weatherCodes[data.code_meteo] || "Conditions inconnues";

return { json: {
ville, temperature: `${data.temperature}°C`, date:
data.heure_releve, message: `À ${ville}, il fait ${data.temperature}°C avec
${description.toLowerCase()}.`, citation: `${data.content}\n de ${data.author}`
} };
```